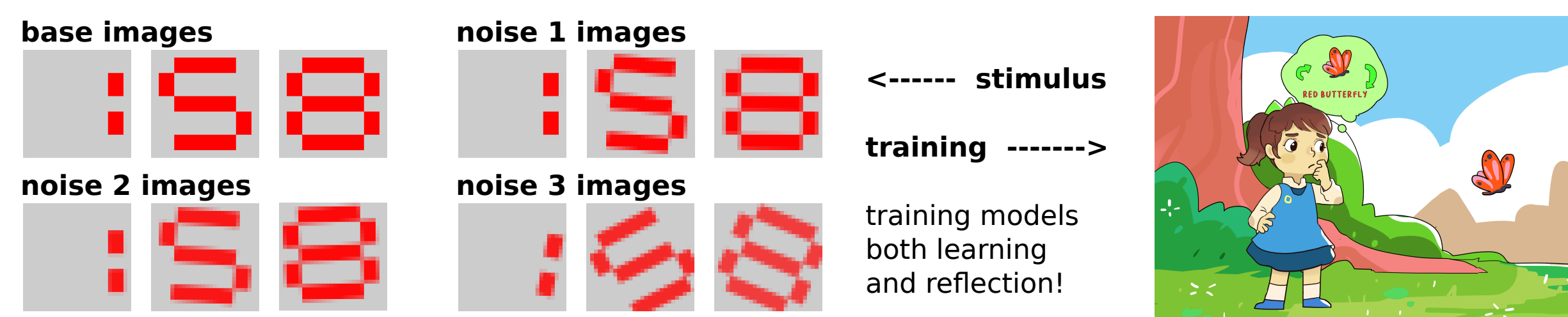


IMAGE, WORD AND THOUGHT

The **iterated learning model (ILM)** simulates the transmission of language from generation to generation in order to explore how the constraints imposed by language transmission facilitate the emergence of language structure.

Despite each learner starting from a blank slate, the presence of a learning bottleneck can lead to the emergence of structured language. Such languages become expressive, compositional, and stable.

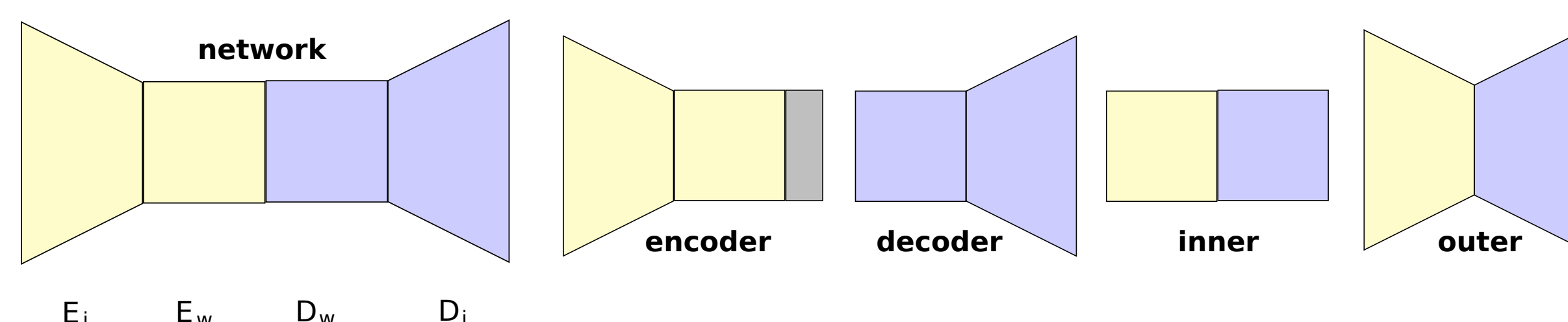
Here, for the first time, the semi-supervised iterated learning model has been applied to a language learning task involving the communication of complex visual meanings: noisy seven-segment display images.



Agents in this model are able to learn and transmit a language that is:

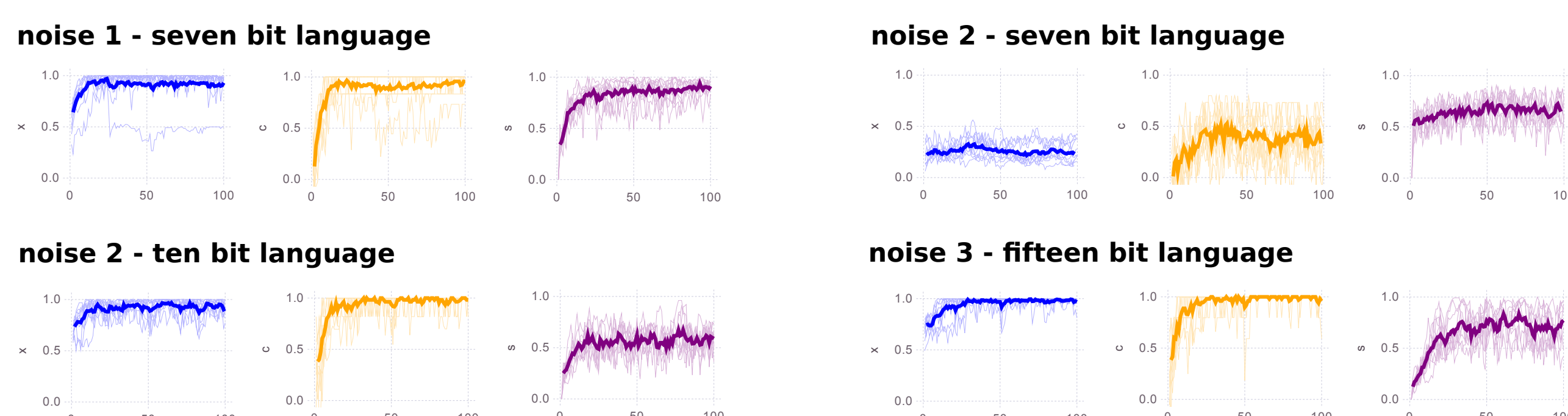
- expressive:** distinct codes are employed for all 128 glyphs;
- compositional:** signal components consistently map to meaning components,
- stable:** the language does not change from generation to generation.

The success of learning relies on a mixture of supervised and unsupervised training; the unsupervised learning is performed on the "outer" network that deals with images and the "inner" network which deals with representations.

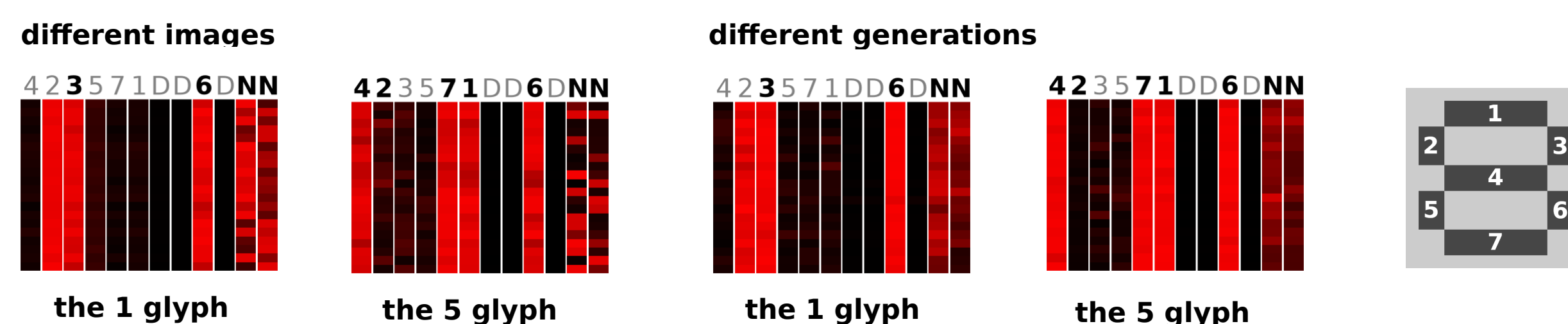


A MORE CHALLENGING TASK FOR THE ILM

An **expressive**, **compositional**, and **stable** language evolves. Thin lines show measures of expressivity, x , compositionality, c , and stability, s , for ten individual 100-generation instantiations. Thick lines show the average values.



Latent activations. These show latent activations for two glyphs. In every case the columns correspond to latent nodes. Some columns have been labelled according to glyph segment; the active segments are labelled in bold



Conclusions: The ILM is a very simple model, but in the semi-supervised version it suggests a striking hypothesis about the origin of language: rather than being wholly concerned with encoding and decoding external utterances from a shared language, the agent must also rely on autoencoder-driven learning of its own "language of thought"

More details: see <https://arxiv.org/abs/2601.02911> and references in that paper. **Contact:** conor.houghton@bristol.ac.uk
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